

Multiple Choice (10 marks)

1. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?
 - (a) $P(A | B) * P(B | C) * P(C | A)$
 - (b) $P(C | A, B) * P(A) * P(B)$
 - (c) $P(A, B | C) * P(C)$
 - (d) $P(A | B, C) * P(B | A, C) * P(C | A, B)$
3. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set
4. MLE estimates are often undesirable because
 - (a) they are biased
 - (b) they have high variance
 - (c) they are not consistent estimators
 - (d) None of the above
5. Averaging the output of multiple decision trees helps ___.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance

True / False (10 marks)

Answer True or False

6. True or False: Supervised learning is regarded as the most innovative technique in Yann LeCun's machine learning cake.
7. True or False: The dimensionality of the null space of the matrix A is 2.
8. True or False: Increasing the number of training examples indefinitely leads to higher variance in the model's predictions.
9. True or False: Predicting the amount of rainfall in a region based on various cues is a supervised learning problem.
10. True or False: RELU activation functions are monotonic, while sigmoid activation functions are not monotonic.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find number of lists present in the given tuple.
12. Write a function to find the largest possible value of k such that k modulo x is y .

End of Paper